

RESEARCH ARTICLE

MAD-CTI: Cyber Threat Intelligence Analysis of the Dark Web Using a Multi-Agent Framework

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ABSTRACT The dark web is a host to illicit activities where hacker forums, blogs, and articles provide significant insights into Cyber Threat Intelligence (CTI) that are frequently unavailable on the surface web. The increasing incidence of security breaches underscores the necessity for advanced CTI solutions to defend against emerging threats. This paper introduces MAD-CTI, a novel multi-agent framework based on Large Language Models (LLM) designed to extract insights from dark web sources. It independently scrapes, analyzes, and classifies content related to vulnerabilities, malware, and hacking, by leveraging a multi-agent architecture to improve efficiency, scalability, and consistency. By utilizing state-of-the-art LLM models and agents, we demonstrate how organizations can adopt this methodology to enhance the accuracy and efficiency of CTI.

INDEX TERMS Cybersecurity defense, cyber threat intelligence, dark web, hack, large language models (LLMs), malware, multi-agent systems (MAS), predictive intelligence, vulnerability.

I. INTRODUCTION

Recent rises in cyber threats have made advanced cyber threat intelligence (CTI) more necessary than ever. From 2021 to 2023, there was a 72% increase in organizational data breaches [1]. These attacks can have damaging effects that go beyond the organizations or individuals themselves and harm users of their software, platform, or data. It is critical that organizations and individuals catch up to modern cyber attack methods with the use of advanced technology to curb this rise in cyber threats.

Some of the most nefarious and damaging attacks tend to arise via the dark web. Over 60% of illicit web activity occurs on the dark web. This is nearly 500 times larger than the surface web [2]. Much of this illegal activity involves the sale of illegal paraphernalia. There is also a significant amount of discussion and sale of ransomware, vulnerabilities, and other malicious pieces of software. To accomplish this, hackers turn to forums or blog posts on the dark web as it allows for anonymity and fewer restrictions than the open/clear web. The dark web hosts a wealth of discussions about potential

future attacks, vulnerabilities in popular software, and/or malware sales, posing significant financial and security risks to enterprises and individuals.

This paper introduces a new tool called **MAD-CTI**, which stands for Multi-Agent Dark Web Cyber Threat Intelligence. MAD-CTI is a novel multi-agent large language model (LLM) tool that scrapes dark web forums for insights on future attacks, vulnerabilities, and malware. It offers organizations and individuals unique information unavailable on the surface web through traditional methods. This novel intel supports proactive threat analysis and the creation of safeguards before attacks, as many vulnerabilities are discussed first on the dark web. While commercial tools report data leaks or breaches on the dark web, little attention is given to monitoring conversations for preemptive intelligence. Recent LLM developments have enabled further exploration in this area.

Our tool is developed using the state-of-the-art GPT-4o mini [3] model from OpenAI, coupled with the multi-agent system framework Microsoft AutoGen [4]. A multi-agent approach to the tool allows dedicated agents to perform each task to ensure efficiency and enhanced performance compared to past iterations of similar tools. These agents

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work together sequentially to scrape dark web pages, decipher human conversations, classify data relevancy, and label the relevant data into the appropriate categories of interest: Hack, Malware, or Vulnerability. We test agents against the CoDA [5] dataset consisting of dark web page extractions relating to the cybersecurity domain.

We will start by exploring existing work in this domain and the limitations we have evolved our tool to address. Then, we will discuss our approach and implementation of the multi-agent architecture. The results of our testing and evaluation immediately follow. Finally, we will discuss potential future work and the overarching conclusion of our work.

II. RELATED WORK

A. SURFACE WEB APPROACHES

Recent advances in cyber threat intelligence have largely focused on extracting information from the surface web. ThreatCrawl is a BERT-based crawler designed to scrape surface websites to determine relevancy to the cybersecurity domain [6]. Once relevancy is established, appropriate keywords are analyzed and the sites are classified as “Malware Used,” “Vulnerability Targeted,” “Broad Information,” or “Tactics, Techniques & Procedures.” The tool achieved a harvest rate of 51.2% (percentage of relevant texts). While the model effectively found relevant information, it fell short in distinguishing different types of cyber threat intelligence due to overlapping content. The tool also only explores clear web data. Some of the approaches can be refined with the utilization of more state-of-the-art technologies and applied to the dark web.

Clairoux-Trepanier, et al. [7] proved the power of using state-of-the-art LLMs like GPT-3.5-turbo by achieving cyber threat classification accuracy of 96.23%, far outperforming previous research in this domain; however, the experiment was only conducted on surface web data as well. We find a similar approach could provide increasingly valuable results when used on dark web data.

B. DARK WEB APPROACHES

The dark web presents unique challenges that require specialized methodologies. While there is an abundance of research focused on the dark web, the technologies typically lag that of the surface web. The work of Kadoguchi, et al. [8] utilized doc2vec and Multi-Level Perceptrons (MLP) to build a dark web threat intelligence tool but suffered when attempting to classify unseen data or when faced with complex context. While their approach is now outdated, it provides a building block and creates a substantial research gap to utilize state-of-the-art technologies to create a more efficient and powerful tool.

Approaches that use more advanced models stick to singular LLM-based agents that continue to lack the context that advanced LLM frameworks can provide. Varghese, et al. [9] build on previous dark web scraping research that

used traditional technologies, such as PageRank [10], TD-IDF [11], SVM, and CNN models [12]. They choose to implement the CyBERT [13] model that is optimized for Named Entity Recognition (NER). They fine-tuned the model on a stronger NER dataset called the Dataset for Named Entity Recognition in Threat Intelligence (DNRTI) [14]. The system utilized the CyBERT model to extract and classify cyber threat entities from a dataset containing conversations from past dark web forums. The tool had promising results by achieving an f1-score of 0.89. This proved to be an improvement on previous NLP tactics at accurately identifying relevant cyber threat-related entities. There is room to build on this research to go beyond just the entity-level. It may be useful to additionally identify the context of the entities by understanding what is being discussed within the forum conversations and how it can be connected to the cyber threat landscape. A more advanced, multi-agent approach would allow us to efficiently extract all this information and derive the appropriate analysis.

Sangher et al. tested a multitude of machine learning and NLP models to build on previous research in this domain and determine the highest-performing model [15]. They found the LSTM model performed the best with a 91.35% accuracy with intent detection and crime classification within dark web hacker forums. Their results show us that a BERT-based approach relied on so heavily may not necessarily be the best approach and that state-of-the-art models need to be further explored. However, the paper was focused on classifying dark web text that only related to social network organizations. While highly performant, the scope of the project is quite limited. We can build on this by creating a more scalable, generalized tool that can provide valuable insights regardless of the entity involved.

C. RESEARCH GAPS

A comprehensive literature survey conducted by Arazzi, et al. [16] showcases how modern NLP techniques are being used to conduct CTI. They discover that work within the dark web CTI domain utilizes an NLP-based approach solely for data collection purposes, entity recognition, or live dark market analysis (e.g. BlackWidow [17]). Per the literature survey and our analysis of related work, it is evident there is little focus on comprehending complex human conversations or writings found within dark web forums, blogs, and/or articles. Where there is research in this domain, it is either limited in scope, lacks scalability, or is not dedicated to the dark web. MAD-CTI is designed to bridge these gaps by employing a multi-agent framework that integrates contextual analysis, improves performance, and enhances scalability.

The summary table at **Table 1** indicates the significant potential to enhance and expand the methodologies of prior research through the implementation of multiple agents capable of conducting an exhaustive analysis of human discourse within darknet websites, and subsequently offering precise cyber threat classifications. Given that dialogues

on the dark web frequently predate actual-world assaults, preemptive measures can be devised based on these insights. With advancements in large-language models and multi-agent LLM technology, it becomes feasible to transcend merely preventing subsequent assaults. Instead, we can advance by acquiring insights that assist end users in discerning potential vulnerabilities within their systems, thus thwarting malicious actors from initiating cyber threats altogether.

III. PROPOSED METHODOLOGY

In the sections that follow, we provide more details of MAD-CTI's architecture and implementation.

A. SYSTEM ARCHITECTURE

Our proposal is to implement a multi-agent architecture that utilizes a state-of-the-art LLM in GPT-4o mini to scrape dark web forums, determine relevancy of information as it pertains to the cybersecurity domain, and classify the context of the conversations to derive actionable cyber threat intelligence.

Unlike a single-agent or purely LLM-based approach, we are able to build an all-inclusive tool that combines the success stories of previous research and leverage parallelism to handle complex tasks efficiently. As depicted in **Fig. 1**, the system includes a specialized agent for each task and communicates its results to the next agent in the sequence. We use Microsoft AutoGen with OpenAI's GPT-4o mini to enable the multi-agent collaboration. Each agent will perform its task individually and pass its findings onto the next LLM agent, communicating via "group chats" and "nested chats."

The workflow is initially fed a dataset of dark web pages. This would come from a separate web scraping agent. However, to test our workflow's performance, we feed it with pre-labeled data from the CoDA database. A User Proxy Agent is used to kickstart the agent chats by instructing them with an initial task. Many blogs/forums are often in a multitude of languages. If needed, a language translator agent will convert the contents to a standard language (in our case, English) for enhanced comprehension between agents. The translated corpus is then passed on to an agent designed to analyze the text's content. It will thoroughly understand the text's semantics and structure to determine the type of information it is dealing with and provide an in-depth analysis of its findings. This analysis is then passed on to the Relevancy Classifier agent that determines whether the text is "Relevant" or "Not Relevant" for gathering cyber threat intelligence. The same analysis is then passed onto a Category Classification Agent that determines the cyber threat category that best fits the context of the text ("Hack," "Malware," "Vulnerability," or "N/A"). Note that these agents do not need to be exposed to the underlying text to form its classification. The analysis and background knowledge passed on from previous agents is enough to form the classification agents' output. The final output is a list of the unique identifiers for each website and the classifications provided by the agents in the form of a CSV file.

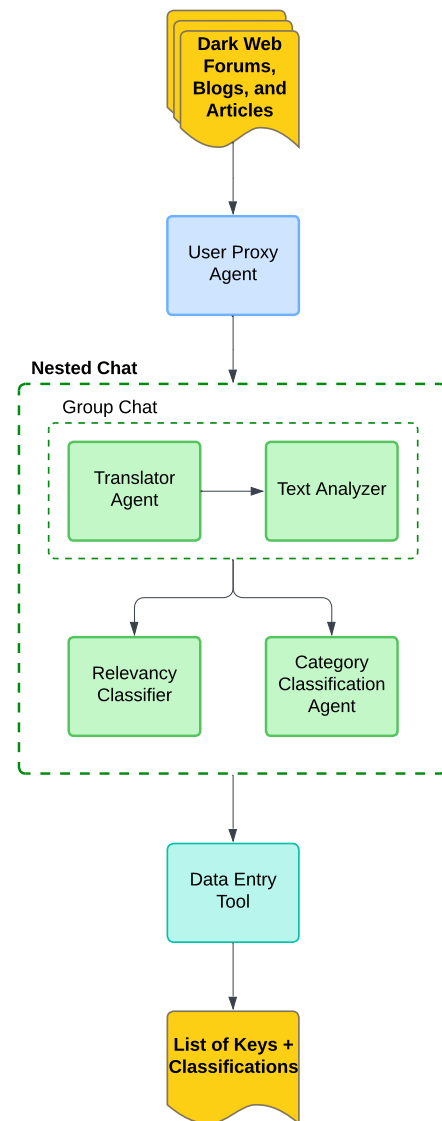


FIGURE 1. Multi-agent workflow.

B. MULTI-AGENT ADVANTAGE

Multi-agent systems (MAS) are systems that utilize multiple computing agents that can interact with one another with little guidance [18]. With the advancements in the artificial intelligence field, this concept can be applied to develop AI/LLM-based agents that can prompt and communicate with one another, with little-to-no human input. We can create a network of conversational agents that operate similarly to that of human workers.

Single agents operate in siloed environments. They are prompted with a task from a human and collect the necessary tools to complete the tasks. Any additional sub-tasks are developed within the same environment. For example, the agent may utilize API calls to external agents to collect information; however, unlike a multi-agent network, no further communication takes place. No memory is retained. In a

TABLE 1. Literature survey comparative analysis.

Feature	P. Kuehn et al.	V. Varghese et al.	K.S. Sangher et al.	M Kadoguchi et al.	V. Clairoux-Trepanier et al.	MAD-CTI
Dark Web Focused	x	✓	✓	✓	x	✓
LLM-Based	✓	✓	x	x	✓	✓
Entity-Agnostic	✓	✓	x	✓	✓	✓
Human Conversational Data	x	✓	✓	x	✓	✓
Multi-Agent System	x	x	x	x	x	✓

multi-agent network, individual agents collect information, retain the memory of this information within a shared memory state, and pass it on to the next agent in the system. The next agent can converse with previous agents without needing to access additional resources of its own or repeat tasks. Sub-tasks can also be created and delegated to the agents who's roles are defined to best handle them. [19]

This approach allows for enhanced performance and efficiency as the agents prompt each other and conduct the appropriate knowledge transfers, requiring very little human intervention. This cohesive environment in which agents generate, prioritize, and execute tasks while auditing one another is what leads to the performance of these models to scale as opposed to single-agent systems [20].

Multi-agent LLMs are a relatively new area in AI, thus applying them to a specialized domain like dark web CTI is not yet widely explored. Our approach solves many common problems faced by similar work when dealing with complex dark web forum data (e.g., comprehension of slang, obfuscation, and multi-language context) by assigning specialized tasks to different agents (e.g., language translation, entity recognition, or threat classification). Existing solutions also typically involve integrating multiple tools for scraping, NLP, and ML separately. A multi-agent LLM-based system could streamline these tasks, providing a unified approach to automatically detect emerging threats, track trends, and/or identify significant conversations with minimal human intervention. Finally, multi-agent architectures can be designed to adapt dynamically based on the nature of the conversation or detected threats, potentially outperforming static models.

IV. UNDERLYING AGENT MODEL

We describe next how MAD-CTI's agents and underlying components work together to realize it.

A. AGENTS

1) DARK WEB SCRAPER

For ethical and accessibility reasons, this study uses the CoDA database, which compiles text from various forums, blogs, and articles that might be hard to access legally. We exclude the scraper agent from our experiment as it is unnecessary for tool evaluation, though it should be included for processing live data. The scraper agent processes onion

links to extract text by using RequestsTor for Tor-proxied connections and BeautifulSoup for scraping. Extracted text is forwarded for further processing. A sample Dark Web Scraper Tool is included in this project's **GitHub repository** that may be easily integrated with an AutoGen agent.

2) TRANSLATOR AGENT

A substantial portion of dark web forums are conducted in English, Russian, German, and other languages. To enhance interpretability for our agents, it is imperative to translate these texts into a standardized language. The Translator Agent is tasked with converting all provided text into English. Consequently, the workflow is structured to accommodate any form of conversational data it encounters.

3) TEXT ANALYSIS AGENT

The Text Analysis Agent is designed to assist the other agents by analyzing the provided text's content and succinctly summarize its findings. The analysis involves the dissection of the conversational structure, narrative consistency, and underlying content of the provided text.

The agent is prompted to identify a clear conversational structure, indicating a forum thread or blog comment. Including these in the analysis helps identify early discussions on potential hacks, malware, or vulnerabilities, which can be useful for preparing vulnerable systems.

To prevent misclassification of advertisements and website front pages as forums or blogs, the agent must analyze text consistency. If a text is a forum front page with links to various threads, it is deemed irrelevant for other agents. The analysis agent informs others about the website's structure and content to focus only on pages offering actionable insights.

The agent should document specific products or services offered, providing insights into hacks or malware related to these products on the dark web. Regardless of features found, an analysis and summary will aid future agents in assessing relevancy and categorization.

4) RELEVANCY CLASSIFIER

The relevancy classifier is designed to process the analysis and summary provided by the Text Analysis Agent to ascertain the relevance of the underlying text. The agent is equipped with sufficient information from the analysis to

make this judgment without the necessity of reviewing the source text directly. The division of this categorization task aims to enhance accuracy; the classifier's exclusive objective is to render a decision while the analysis agent is tasked solely with examining the text.

In instances where the content lacks specific product-/services (commonly downloadable files), a distinct conversational structure, or a consistent narrative, the Relevancy Classifier Agent will designate the key associated with the text as "Not Relevant." Conversely, if the text resembles a forum thread, blog post, or article that potentially offers valuable insights into possible hacks, malware, or vulnerabilities, it will assign the label "Relevant." The key corresponding to the text (or the onion link if the application is deployed on a live webpage) alongside the label assigned by the agent is documented in a CSV file.

5) CATEGORY CLASSIFICATION AGENT

The Category Classification Agent receives and processes the same analysis generated by the Text Analysis Agent. Its primary function is to classify the underlying text as a potential hack/attack, malware, or vulnerability since these are 3 of the most common types of cyber threats. Should the Relevancy Classifier Agent assess the text as irrelevant, the Category Classification Agent is programmed to automatically assign the category as "N/A."

The agent is fed the same specifications we had set for ourselves when labeling the CoDA dataset. In order to be classified as one of the 3 labels, it must fit the following descriptions:

- 1) **Hack:** The text details hacking methods or techniques. This typically involves account hacking, password cracking, DDOS attacks, phishing, or SQL injection.
- 2) **Malware:** The text discusses malware in software, systems, or programs. This may typically be ransomware, viruses, spyware, trojans, or keyloggers.
- 3) **Vulnerability:** The text describes a vulnerability that can be exploited in a system. This usually involves descriptions of bugs and exploits within software, organizations, or computers that can be taken advantage of.

The final label is added to a separate column in the same CSV file the Relevancy Classifier Agent entered its data in.

B. MICROSOFT AUTOGEN

We used Microsoft AutoGen to create a workflow enabling agent communication. AutoGen is a multi-agent framework that supports collaboration among LLM-based agents with built-in "Assistant," "Conversable," and "User Proxy" agent frameworks. These are configured with LLM settings, customizable parameters, tools, and prompts to perform actions and communicate autonomously. AutoGen is the foundation of our automated, cohesive multi-agent environment.

C. OPENAI GPT-4O MINI

We chose OpenAI's GPT-4o mini for its high compatibility with AutoGen as the framework was designed to be most effective using OpenAI models. We opted for the more efficient GPT-4o mini over the parent model GPT-4o after testing both models against half the dataset. GPT-4o mini matched GPT-4o's performance metrics but was far more cost-effective, prompting us to use it for testing. When compared to Anthropic's equivalent lightweight LLM Claude 3.5 Haiku, GPT-4o mini far outperformed in both performance and cost. Refer to **Appendix C** for full testing results.

With the rapid evolution of LLMs, increasingly advanced models will soon become more affordable. AutoGen's modularity and adaptability lets users easily switch LLMs, allowing experimentation even after GPT-4o mini is outdated. For our research, GPT-4o mini was the most accessible and sufficient for demonstrating our tool's superiority. The experiment can be replicated using more expensive options, such as GPT-4o [21] or Claude 3 Opus [22].

V. TESTING AND EVALUATION OF MAD-CTI

A. DEVELOPMENT OF TESTS

Before running the workflow on live websites, we fed the workflow 650 pieces of cyber threat-related texts from the CoDA database. Only cyber threat-related texts were extracted from the database and each text was manually annotated with the appropriate relevancy and category labels (refer to **Appendix A** for annotation rationale). Then, through an iterative approach of testing and evaluating each individual agent, the prompts were fine-tuned to provide the agents with increasingly specific instructions on what to look for within the texts to perform their assigned task(s).

Initial trials of the Relevancy Agent resulted in too many texts being marked as "Relevant." Upon analysis, many were merely advertisements or website front pages, lacking valuable insights. We refined the agent's prompts with more specific criteria and negative samples, repeating the process until achieving a harvest rate of 45.15%, aligning closer to the actual rate. Enhanced communication with the Text Analysis Agent was also implemented.

The Category Classification Agent improved through development by observing LLM and Text Analysis Agent interactions. This helped refine prompts for clarity in categorizing text as "Hack," "Malware," or "Vulnerability." Examples of hacks and malware were provided to aid differentiation, enhancing workflow performance.

Overarching agents use a group chat with multiple assistant agents through Microsoft AutoGen. Once developed and prompts refined, each agent joins a nested group chat, enabling "agents of agents" to interact. A user starts the initial chat and sets termination instructions, initiating the workflow. We observe agent interactions to identify and address any information loss for prompt improvement.

B. RESULTS OF TESTING

As mentioned, all classifications developed by the agents is inputted into a CSV file. We compared the results of this CSV file with the pre-labeled dataset and achieved the results found in **Table 2**.

TABLE 2. Relevancy label performance.

Label	Precision	Recall	F1-Score
Relevant	0.6604	0.5993	0.6284
Not Relevant	0.6937	0.7465	0.7191
Weighted Average	0.6787	0.6801	0.6782
Accuracy	0.6801		

Note: Boldface indicates best performance.

We observe the agents performed well at identifying texts that are “Not Relevant” based on our specifications (F1-score of 0.72) but were slightly worse when it came to texts that should have been “Relevant” (F1-score of 0.63). The Precision and Recall scores for the “Relevant” label tells us the workflow does a good job at correctly identifying relevant texts when it makes that distinction. However, it is still prone to be conservative by making more texts as “Not Relevant” per the Recall scores. Ideally, the agents should actually be more sensitive by identifying more texts as relevant as an end user would theoretically rather catch as many insights as possible due to the nature of what is being analyzed here. However, we must also have a trade-off between accuracy and sensitivity. End users also do not need to be overloaded with inaccurate information. The agents do a good job of balancing this tradeoff with a weighted average F1-score of 0.68 (calculated using Support values as weights) and total accuracy of 68.01%.

The workflow showed significant variability in predicting the threat categories. As seen in **Table 3**, it particularly struggled with identifying vulnerabilities. This is likely due to the sample size as there were not enough texts to be labeled as “Vulnerability.” It had a good balance of Precision and Recall scores for the other labels to achieve a weighted average F1-score of 0.63 and overall accuracy of 63.52%. However, since the “N/A” label is directly tied to the “Not Relevant” label, the weighted average F1-score for the rest of the labels is 0.53, with “Malware” performing the best with an F1-score of 0.58.

TABLE 3. Category label performance.

Label	Precision	Recall	F1-Score
Hack	0.5487	0.5194	0.5337
Malware	0.6792	0.5070	0.5806
Vulnerability	0.1765	0.2000	0.1875
N/A	0.6937	0.7465	0.7191
Weighted Average	0.6340	0.6352	0.6326
Accuracy	0.6352		

Note: Boldface indicates best performance. We exclude “N/A” label from this as it is directly tied to the “Not Relevant” label from **Table 2**.

VI. EVALUATION AGAINST PRIOR APPROACHES

Next, we compare the performance of MAD-CTI against several approaches in prior work.

A. MAD-CTI VS. DARKBERT

DarkBERT is a BERT-based language model trained on dark web forum data. While DarkBERT provided promising results and did a good job categorizing highly complex human conversation, we sought out to find areas to improve on through alternative LLMs/model approaches. DarkBERT was evaluated against both the DUTA and CoDA datasets.

It can be observed in **Table 4** the model performed much worse at detecting relevancy compared to our multi-agent approach with the use of GPT-4o mini. When comparing our “Relevant” label F1-score with DarkBERT’s raw “noteworthy thread detection” results, we observe an 18.90% increase in performance. Our approach requires no prior preprocessing of the data since the agents can perform any processing needed on their own and still performs significantly better than both the raw and preprocessed DarkBERT models.

TABLE 4. DarkBERT noteworthy thread detection performance.

Input	Model	Precision	Recall	F1 score
Raw	BERT _{cased}	0.5509	0.1991	0.2690
	BERT _{uncased}	0.5234	0.2349	0.2851
	RoBERTa	0.2897	0.1789	0.2138
	DarkBERT _{raw}	0.7593	0.4308	0.5285
Preprocessed	BERT _{cased}	0.6143	0.2048	0.2881
	BERT _{uncased}	0.4546	0.2152	0.2616
	RoBERTa	0.2904	0.1527	0.1871
	DarkBERT _{preproc}	0.7244	0.4513	0.5417

Note: Boldface indicates best performance.

B. MAD-CTI VS. THREATCRAWL

Much of our experiment rationale was motivated by ThreatCrawl. ThreatCrawl scrapes surface web sites and provides similar relevancy and category labels. While ThreatCrawl aims to provide similar insights as us, it was limited in its scope to the surface web while we focused on dark web data. However, it is still helpful to showcase the improvements our approach provides and how it can possibly be implemented on the surface web as well (with likely higher success as surface web data is significantly less complex).

ThreatCrawl had similar classifications to MAD-CTI by classifying each “Relevant” web scraped text as either “Malware” or “Vulnerability”. The tool was run using CySecBERT [23], BERT [24], and SBERT [25]. Their results can be seen in **Table 5**. While the BERT-based models performed well when it came to discerning relevancy of texts, it struggled at accurately identifying the category that best described the text. When compared to our approach, it is clear that the multi-agent approach using GPT-4o mini outperforms individual BERT-based approaches for similar tasks.

TABLE 5. ThreatCrawl performance comparison of different BERT models.

	CySecBERT			BERT			SBERT		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
<i>Rel.</i>	0.62	0.99	0.76	0.77	0.98	0.86	0.73	0.98	0.83
<i>Malw.</i>	0.27	0.05	0.08	0.43	0.17	0.22	0.39	0.16	0.22
<i>Vulns.</i>	0.39	0.97	0.49	0.41	0.91	0.56	0.33	0.92	0.40

C. SINGLE AGENT VS. MULTI-AGENT APPROACH

In order to demonstrate the efficacy of the multi-agent architecture, we conducted an experiment utilizing a singular agent to execute the identical tasks. A comprehensive prompt encompassing all the tasks described in our principal experiment was provided to this singular agent. (see **Appendix B** for full prompt). We see in **Table 6** how the singular agent struggled to accurately output the correct labels compared to when we split the tasks up between multiple agents.

TABLE 6. Single-agent relevancy label performance.

Label	Precision	Recall	F1-Score
Relevant	0.5470	0.9932	0.7055
Not Relevant	0.9829	0.3230	0.4863
Weighted Average Accuracy	0.7861	0.6256	0.5852

Note: Boldface indicates best performance.

The standard, single-agent approach yielded an overall accuracy for the relevancy label of 62.56% compared to 68.01% from the multi-agent approach, an 8.01% decrease. Additionally, we can observe an interesting pattern of a high Recall but low Precision score for the “Relevant” label. This tells us the single-agent approach tends to label more texts as “Relevant” than it should. While this catches most of the relevant tasks, it does not do so accurately. Conversely, we see a high Precision but low Recall for the “Not Relevant” label, indicating that when the agent does decide the text is “Not Relevant,” it does so at a high accuracy. Unfortunately, it is hesitant to do so as we see many irrelevant texts being marked as “Relevant.” This lack of balance between sensitivity and accuracy led to a weighted average F1-score drop of 13.71% compared to the multi-agent workflow.

Table 7 shows how the single-agent category labeling accuracy dropped to 54.89%. This is a 13.59% reduction from the multi-agent model. This also marks a drop-off in accuracy between tasks of 12.26% compared to just a 6.60% drop-off between tasks in the multi-agent approach. The individual agent also performed slightly better when predicting “Hack” or “Vulnerability” as the text’s category label. However, it suffered significantly with “Malware” and “N/A” labels.

This overall performance drop-off is expected since one agent has to complete multiple tasks at once. There is no room for specialization. Meanwhile, multiple agents can be prompted to be specialized in a specific task. This allows for the breakdown of complex tasks into smaller, simpler tasks;

TABLE 7. Single-agent category label performance.

Label	Precision	Recall	F1-Score
Hack	0.4444	0.9512	0.6058
Malware	0.4471	0.6230	0.5205
Vulnerability	0.4444	0.3077	0.3636
N/A	0.9813	0.3344	0.4988
Weighted Average Accuracy	0.7501	0.5489	0.5298

Note: Boldface indicates best performance. We exclude “N/A” label from this as it is directly tied to the “Not Relevant” label from **Table 6**.

hence, why we see a significant increase in performance in most tasks through our multi-agent approach.

VII. FUTURE WORK

MAD-CTI constitutes a promising foundation for an advanced cyber threat intelligence tool targeting the dark web. We have engineered a tool capable of accurately interpreting intricate human dialogues, facilitating the requisite classification and analysis of cyber threats, thus aiding end users in initiating suitable remediation actions within their systems. Nevertheless, the extant tool is designed to demonstrate the efficacy of multiple agents operating collaboratively to derive actionable insights from a highly complex domain. Consequently, opportunities exist for the integration of additional agents to develop a more sophisticated tool. This potential expansion may encompass:

- **Advanced Crawler Agent:** As previously mentioned, our scraper agent was not utilized due to accessibility issues with dark web forums along with ethical considerations. However, the agent is still a crucial part of the final workflow and can be improved. The current iteration is a simple scraping agent that connect to the Tor proxy and extracts text using BeautifulSoup. This is only helpful in analyzing one website and is not fully equipped with the protocols needed to safely navigate the dark web. Thaar Asman and Haider Abd detail the difficulty to legally and ethically scrape dark web data [26]. The need for dedicated agents that handle the complex and decentralized nature of dark web data further reinforces the benefits a multi-agent LLM-based approach. Additionally, enhanced insights can be gathered by automatically crawling through links within a provided website. This crawler would enhance the automation of our workflow as the tool would crawl through a multitude of forum posts anonymously and structurally on its own with little-to-no human intervention.
- **Named Entity Recognition (NER) Agent:** This agent is responsible for extracting the specific entity being targeted in the discussion being analyzed. For example, much of the CoDA dataset included conversations surrounding Facebook account hacking. A company/individual would find the tool’s insights most impactful if

they are only alerted when it is their systems at risk. This opens the door for increased specialization of our tool.

- **Risk Analysis Agent:** An agent that can calculate a risk score based on the information provided by other agents would provide a valuable metric to companies/individuals on the magnitude of the impact a specific threat may pose. For example, a general workaround to a system that is already known by a user and is being worked on would not be as “risky” as a novel malware. Discussion of how a hacker group plans to attack a company’s system would hold a high risk rating as a company would need to immediately patch their systems and preemptively protect themselves from these future attacks. Risk would be dependent on sensitivity, novelty, and timeline of the information. The Risk Analysis Agent would help our tool reach its fullest potential by providing a quick and easy metric to monitor, all with no human intervention needed.

Investigating the enhancement of workflow performance through the employment of multiple large language models (LLMs) would be advantageous. For instance, DarkBERT has demonstrated commendable named entity recognition (NER) capabilities and can serve as an alternative to GPT-4o mini for the NER agent due to its fine-tuning specifically on dark web data. Additionally, state-of-the-art LLMs like OpenAI o1 [27] or Claude 3.5 Sonnet [28] provide highly advanced (albeit slow) reasoning capabilities that could add significant enhancements to our tool’s accuracy as they have already shown to enhance performance on complex tasks, such as ours. Unfortunately, due to current rate limitations and resource constraints, these models were unable to be utilized in our experiments.

Additionally, expanding the scope of each agent’s classification criteria could increase accuracy as each agent may have more freedom in what it classifies a text as. For example, we can split the “Malware” category to “Ransomware,” “Spyware,” and “Virus.” Rather than having to occasionally “guess” and generalize the text into a forced category, we may find accuracy increasing when we provide the agents with more category options. Fine-tuning the agents’ knowledge base with cybersecurity datasets such as MITRE ATT&CK may prepare them for increasingly specific classification tasks [29].

VIII. CONCLUSION

In this paper, we introduced MAD-CTI (Multi-Agent Dark Web Cyber Threat Intelligence), a novel multi-agent large language model framework designed to extract actionable insights from dark web forums, blogs, and articles. Our research addresses critical gaps in existing cyber threat intelligence approaches by leveraging a sophisticated multi-agent architecture that can comprehensively analyze complex dark web conversations with little-to-no human intervention.

The use of GPT-4o mini and Microsoft AutoGen allows us to break down complex CTI-related tasks into specialized

agents that collaborate with one another to develop a more performant overall system. By leveraging dark web data over surface web data, our tool allows for a nuanced understanding of forum discussions, potentially enabling proactive threat detection and mitigation for end users. Our approach demonstrated effectiveness by achieving an overall accuracy of 68.01% for relevancy classification and 63.52% for threat categorization, along with average F1-scores of 0.6737 and 0.5052, respectively. This marked a significant improvement over similar models like DarkBERT and ThreatCrawl, where our approach provides an 18.90% increase in performance for relevancy classification compared to DarkBERT and a sizeable improvement in categorical classification compared to ThreatCrawl.

We also highlight the advantages of a specialized, task-specific multi-agent system over traditional single-agent approaches. Our multi-agent approach showcases an 11.55% increase in relevancy classification performance and 13.59% increase in category labeling compared to a standard single-agent approach where only one prompt is used to complete multiple tasks at once.

The research identifies several promising directions for future investigation. Potential improvements include the development of a sophisticated crawler agent, the implementation of a named entity recognition agent, and the creation of a risk analysis agent. Furthermore, the examination of multiple large language models may enhance the tool’s performance and precision.

The current implementation demonstrates a promising proof of concept, highlighting the substantial potential of multi-agent architectures within cybersecurity. As cyber threats persist in increasing complexity, tools such as MAD-CTI signify a crucial advancement toward the development of more adaptive and intelligent threat intelligence systems.

APPENDIX A DATA ANNOTATION

To measure our agentic workflow’s performance, a pre-labeled dataset was required to evaluate the labels outputted by the Relevancy and Category Classification Agents. Unfortunately, the CoDA database is not labeled past a highly general category label (e.g. “Hacking,” “Gambling,” “Financial,” etc.). Hence, we had to manually label all 650 cyberthreat-related texts as either “Relevant” or “Irrelevant” and “Hack,” “Malware,” “Vulnerability,” or “N/A.” Consistency in labeling these texts was crucial. We assigned strict guidelines for each text to be considered “Relevant” and what category best fits it:

- **Relevant:** The text should be that of a forum post, blog, or news article. There needs to be a clear conversation between multiple people to be considered a forum post. A specific product/service being sold may provide insight into novel hacks/malware and is considered relevant.
 - **Hack:** A specific hack in the form of a guide, service, etc.

- **Malware:** Discussion or sale of a specific software or virus that can take control of a system or breach its data.
- **Vulnerability:** An exploit of a system, website, etc. that can be taken advantage of by bad actors.
- **Not Relevant:** Any broken/incoherent text is not relevant. Text primarily made up of ads, is a front page for a website, and/or has no consistent narrative is irrelevant.
 - **N/A:** All text that is labeled as “Not Relevant” should also have the sub-category set to “N/A”.

After carefully analyzing each text and applying our label to them based on the rationale above, these same guidelines were provided to the AI agents within our workflow.

APPENDIX B SINGLE-AGENT VS. MULTI-AGENT APPROACH

We observe a sample in which the single-agent approach incorrectly classifies an underlying text. Fig. 2 shows a sample prompt from the CoDA database. It is evident this piece of text is simply an extraction of a front page of a website or forum. This would fall under the “Not Relevant” category based on the rationale we have previously provided.

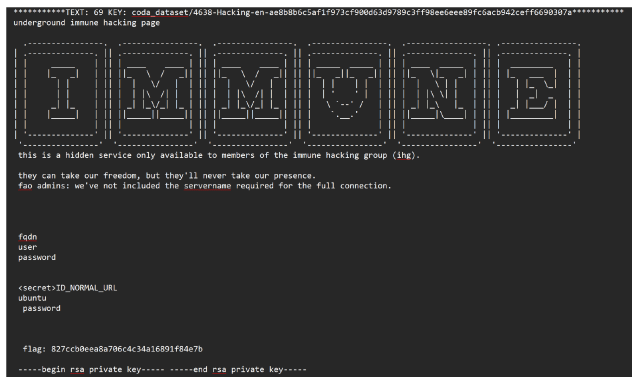


FIGURE 2. Sample text used to compare MAD-CTI to single-agent approach.

Fig. 3 shows the appropriate analysis and correct labeling from the MAD-CTI multi-agent tool. Meanwhile, we see in Fig. 4 how a singular agent incorrectly marks this text as “Relevant” and a “Hack.” The single agent does not break the problem down into multiple tasks and jumps directly to an answer. We do not see a glimpse into its rationale under-the-hood and measure whether it is correctly approaching the task.

The prompt fed to the single agent can be found in Fig. 5. Due to the complexity of the tasks, the agent is inputted a rather large prompt where it is told to complete all the tasks (that would have otherwise been split up into multiple agents) on its own and this leads to suffering performance. Hence why the multi-agent approach has proven to be far superior as it gradually arrives to its solution by splitting the task at

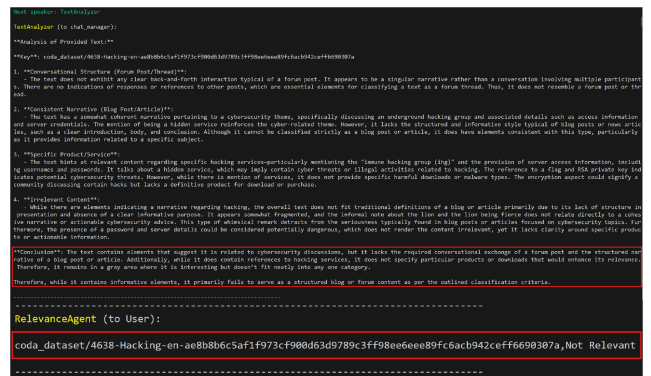


FIGURE 3. MAD-CTI output for sample text from Fig. 2.

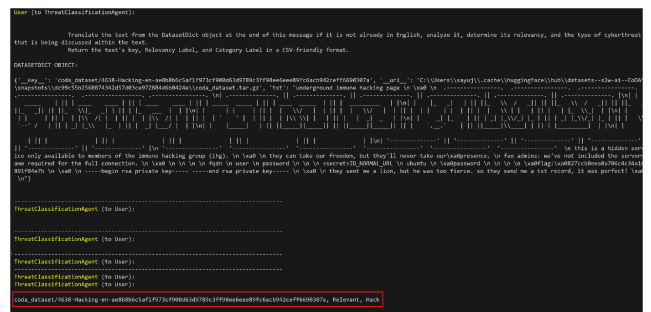


FIGURE 4. Single-agent output for sample text from Fig. 2.

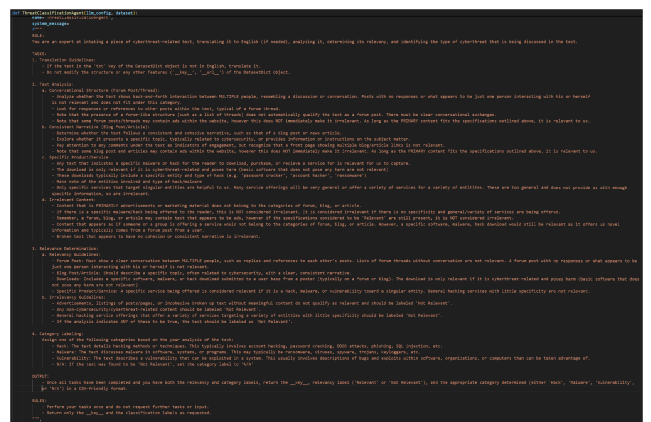


FIGURE 5. Prompt inputted to singular agent.

hand into multiple smaller tasks and communicates with other agents to achieve a more accurate result.

APPENDIX C GPT-4O MINI VS. OTHER MODELS

When deciding what model to use for our testing, we first tested our workflow using OpenAI’s GPT-4o API to explore whether there was an increase in accuracy compared to GPT-4o mini. We found the model ran significantly longer and was much more computationally and financially expensive compared to GPT-4o mini, while yielding near identical results, even being outperformed in overall accuracy and with labeling “Not Relevant” texts. The results after testing each

TABLE 8. GPT-4o vs. GPT-4o mini.

Label	GPT-4o			GPT-4o mini		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Relevant	0.5691	0.7630	0.6519	0.6493	0.6397	0.6444
Not Relevant	0.7762	0.5873	0.6687	0.7435	0.7513	0.7474
Weighted Average Accuracy	0.6899	0.6605	0.6617	0.7040	0.7046	0.7043
	0.6605			0.7046		
Hack	0.5385	0.7000	0.6087	0.5300	0.5761	0.5521
Malware	0.5490	0.7568	0.6364	0.6667	0.4324	0.5246
Vulnerability	0.3333	0.5714	0.4211	0.2000	0.2857	0.2353
N/A	0.7762	0.5873	0.6687	0.7435	0.7513	0.7474
Weighted Average Accuracy	0.6743	0.6378	0.6429	0.6626	0.6554	0.6557
	0.6378			0.6554		

Note: Boldface indicates best performance. We exclude "N/A" label from this as it is directly tied to the "Not Relevant" label.

TABLE 9. Claude 3.5 Haiku vs. GPT-4o mini.

Label	Claude 3.5 Haiku			GPT-4o mini		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Relevant	0.4600	0.8984	0.6085	0.6493	0.6397	0.6444
Not Relevant	0.7937	0.2703	0.4032	0.7435	0.7513	0.7474
Weighted Average Accuracy	0.6572	0.5272	0.4872	0.7040	0.7046	0.7043
	0.5272			0.7046		
Hack	0.4000	0.8276	0.5393	0.5300	0.5761	0.5521
Malware	0.5686	0.8286	0.6744	0.6667	0.4324	0.5246
Vulnerability	0.1053	0.3333	0.1600	0.2000	0.2857	0.2353
N/A	0.7937	0.2703	0.4032	0.7435	0.7513	0.7474
Weighted Average Accuracy	0.6459	0.4888	0.4667	0.6626	0.6554	0.6557
	0.4888			0.6554		

Note: Boldface indicates best performance. We exclude "N/A" label from this as it is directly tied to the "Not Relevant" label.

LLM on a subset of about half the dataset can be found in **Table 8**.

Additionally, we compared GPT-4o mini to Anthropic's equivalent cost-effective, efficiency-focused model Claude 3.5 Haiku. When tested against the same data subset as the experiment in **Table 8**, we found the model struggled heavily with Microsoft AutoGen's framework as the framework is most compatible with OpenAI models. This led to highly inaccurate results and significant human intervention to manually clean output data that was otherwise seamless with GPT-4o mini. We observe a 25.18% dropoff in overall relevancy classification accuracy and 25.42% decrease in category labeling accuracy, with the most significant factor being Claude 3.5 Haiku's substantial amount of false positives. The weighted average Precision, Recall, and F1-Scores all performed superior when using GPT-4o mini.

ETHICAL CONSIDERATIONS

The dark web is an unregulated area of the internet full of illicit data. It was essential to consider the ethical and legal implications of all dark web activities involved in our experiment. We carefully adhered to the DICE-E [30] framework to guide our experiment through the safe identification, collection, and evaluation of the dark web data we used for our model. The DICE-E framework offers structured plans on ensuring privacy and safety of both the researchers and darknet forum participants. The CoDA database also strictly adheres to the ACM Code of Ethics [31], ensuring all forum participant information is appropriately masked for the safety and privacy of all users involved.

In order to access this dataset, we had to apply for use and accept all terms of use associated with it. Our approach prioritized research integrity and the responsible use of dark web intelligence, focusing solely on gathering insights to enhance cybersecurity defenses.

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